

Performance Testing

Date	2 October 2026
Team ID	SWTID-2026-6275
Team Name	Cipher
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Project Name	Legal Ease: AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust (Python-based load testing tool)
Type of Testing	Load Testing, Stress Testing and Spike Testing
Target Module / API	Document Generation API (POST /api/generate-document)
Test Environment	Local development server (localhost)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal load: login, browse templates, generate document	10	120	Avg < 2 s, 0% errors
2	Peak load: concurrent rental agreement and NDA drafts	50	300	Avg < 2 s, error rate < 1%
3	Stress test: gradual ramp-up to find breaking point	100	300	Stable; degrades gracefully
4	Spike test: sudden jump from 10 to 80 users	80	180	Recovers to normal after spike

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Within target at 50 users
2	Response Time (Max)	< 5 seconds	3.85 seconds	Pass	Highest value seen during spike test
3	Throughput (Req/sec)	> 20 req/sec	34.6 req/sec	Pass	Stable throughout the run
4	Error Rate	< 1%	0.4%	Pass	Few timeouts at peak of stress test
5	CPU Utilization	< 80%	68%	Pass	Peak CPU during stress test
6	Memory Utilization	< 80%	61%	Pass	No memory leak observed

Step 4: Observations & Analysis

Key Findings:

The document generation API met all target values in the Locust load, stress and spike tests. Average response time was 1.42 seconds at 50 concurrent users, with a 0.4% error rate and throughput of 34.6 requests/sec. The system stayed stable up to about 80 users; beyond that, response times rose gradually.

Bottlenecks Identified:

The slowest step was the AI draft generation call, which took most of the response time. Under the stress test (100 users), a few requests timed out and CPU usage rose to about 68%. Repeated database reads for the same legal templates added extra delay.

Optimization Steps Taken:

Cached frequently used legal templates to cut repeated database queries, added database connection pooling, limited the maximum size of each generation request, and set a timeout with a retry for the AI service call. After these changes, the average response time dropped and the error rate stayed below 1%.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1: Locust statistics table (request count, response times, failures)

Screenshot 2: Locust charts (total requests per second, response times, number of users)